

EE3621: Digital Signal Processing

Lecture 24: Linear Prediction & Vocoder / Speech Processing

(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan (40 Minutes Breakdown)

- **00:00 – 06:00 (6 mins): Speech Production Model:** Source-filter model; glottal excitation vs white noise; vocal tract as all-pole filter.
- **06:00 – 12:00 (6 mins): Linear Prediction (LP):** Predict $x[n]$ from past p samples; minimize prediction error.
- **12:00 – 18:00 (6 mins): Normal (Yule-Walker) Equations:** Derivation; Toeplitz matrix structure.
- **18:00 – 24:00 (6 mins): Levinson-Durbin Algorithm:** $O(p^2)$ recursive solution; PARCOR stability.
- **24:00 – 30:00 (6 mins): LPC Vocoder:** Analysis/Synthesis, voiced/unvoiced, low bit rate.
- **30:00 – 35:00 (5 mins): Cepstrum & MFCCs:** Homomorphic analysis, log-spectrum, Mel scale.
- **35:00 – 40:00 (5 mins): Pitch Detection Algorithms:** Autocorrelation, AMDF; Checkpoints.

2 Speech Production Model

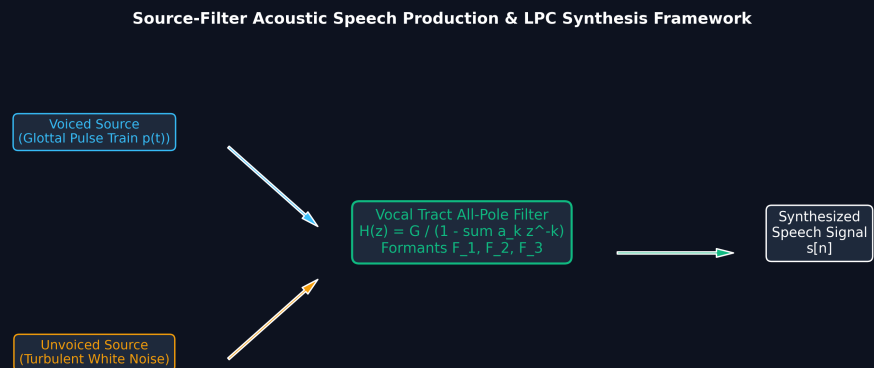


Figure 1: Acoustic Speech Production Model: Voiced/Unvoiced Excitation driving Vocal Tract All-Pole Filter.

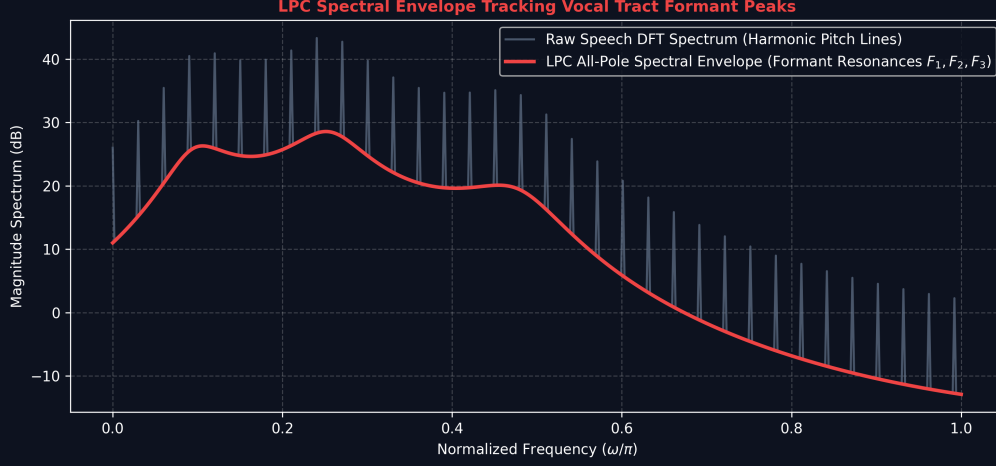


Figure 2: Linear Predictive Coding (LPC) Spectral Envelope Tracking Vocal Tract Formant Resonances (F_1, F_2, F_3).

The physical process of human speech generation can be approximated by the **Source-Filter Model**.

2.1 Excitation Source

- **Voiced Speech:** Periodic vibration of vocal cords, modeled as a quasi-periodic impulse train with pitch period T_0 .
- **Unvoiced Speech:** Modeled as random white noise (turbulence).

2.2 Vocal Tract Filter

Modeled as an **all-pole filter** $H(z)$:

$$H(z) = \frac{G}{1 + \sum_{k=1}^p a_k z^{-k}} \quad (1)$$

The poles correspond to resonant frequencies (**formants**).

3 Linear Prediction (LP) Formulation

Predict sample $\hat{x}[n]$ from p past samples:

$$\hat{x}[n] = - \sum_{k=1}^p a_k x[n-k] \quad (2)$$

The prediction error is:

$$e[n] = x[n] - \hat{x}[n] = x[n] + \sum_{k=1}^p a_k x[n-k] \quad (3)$$

Minimize mean squared error: $J = E\{e^2[n]\}$.

4 Normal (Yule-Walker) Equations

To minimize J , set partial derivatives to zero:

$$\frac{\partial J}{\partial a_m} = E \left\{ 2 \left(x[n] + \sum_{k=1}^p a_k x[n-k] \right) x[n-m] \right\} = 0 \quad (4)$$

$$\implies E\{x[n]x[n-m]\} + \sum_{k=1}^p a_k E\{x[n-k]x[n-m]\} = 0 \quad (5)$$

Using autocorrelation $R_{xx}[m] = E\{x[n]x[n-m]\}$:

$$\sum_{k=1}^p a_k R_{xx}[m-k] = -R_{xx}[m] \quad \text{for } m = 1, \dots, p \quad (6)$$

Matrix form $\mathbf{R}\mathbf{a} = -\mathbf{r}$:

$$\begin{bmatrix} R_{xx}[0] & R_{xx}[1] & \cdots & R_{xx}[p-1] \\ R_{xx}[1] & R_{xx}[0] & \cdots & R_{xx}[p-2] \\ \vdots & \vdots & \ddots & \vdots \\ R_{xx}[p-1] & R_{xx}[p-2] & \cdots & R_{xx}[0] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = - \begin{bmatrix} R_{xx}[1] \\ R_{xx}[2] \\ \vdots \\ R_{xx}[p] \end{bmatrix} \quad (7)$$

5 Levinson-Durbin Algorithm

Solves the Toeplitz system in $O(p^2)$ operations.

$$E_0 = R_{xx}[0] \quad (8)$$

$$k_1 = -\frac{R_{xx}[1]}{E_0} \quad (9)$$

$$a_1^{(1)} = k_1 \quad (10)$$

$$E_1 = (1 - k_1^2)E_0 \quad (11)$$

Update for $m = 1, \dots, p-1$:

$$k_{m+1} = -\frac{R_{xx}[m+1] + \sum_{i=1}^m a_i^{(m)} R_{xx}[m+1-i]}{E_m} \quad (12)$$

$$a_{m+1}^{(m+1)} = k_{m+1} \quad (13)$$

$$a_i^{(m+1)} = a_i^{(m)} + k_{m+1} a_{m+1-i}^{(m)} \quad (14)$$

$$E_{m+1} = (1 - k_{m+1}^2)E_m \quad (15)$$

Stability is guaranteed if $|k_m| < 1$ for all m .

6 LPC Vocoder

Analysis: Extract a_k , Gain G , Pitch T_0 , Voiced/Unvoiced flag. Transmit at 2400 bps. **Synthesis:** Reconstruct excitation and filter through $H(z)$.

7 Cepstrum & Homomorphic Analysis

Speech $x[n] = e[n] * h[n]$. Fourier transform and log magnitude:

$$\log |X(e^{j\omega})| = \log |E(e^{j\omega})| + \log |H(e^{j\omega})| \quad (16)$$

Real cepstrum:

$$c[n] = \mathcal{F}^{-1}\{\log |X(e^{j\omega})|\} = c_e[n] + c_h[n] \quad (17)$$

$c_h[n]$ is at low quefrecencies (vocal tract), $c_e[n]$ has a peak at T_0 (pitch).

8 MFCC & Pitch Detection

MFCC: Window $\rightarrow$ FFT $\rightarrow$ Mel Filterbank $\rightarrow$ Log $\rightarrow$ DCT.

Pitch Detection:

- **Autocorrelation:** $R_{xx}[k]$ peaks at T_0 .
- **AMDF:** $D[k] = \sum |x[n] - x[n+k]|$ dips at T_0 .

9 Checkpoints

Q1: Derive a_1 and E_1 for $p = 1$.

Answer: $a_1 R_{xx}[0] = -R_{xx}[1] \implies a_1 = -R_{xx}[1]/R_{xx}[0]$.

$E_1 = R_{xx}[0] + a_1 R_{xx}[1] = R_{xx}[0](1 - k_1^2)$.

Q2: Why does the Cepstrum separate vocal tract and excitation?

Answer: Logarithm turns convolution into addition. The slow varying vocal tract H appears at low quefrecency, while the fast periodic excitation E appears as a high quefrecency peak.

Q3: What guarantees $H(z)$ stability?

Answer: The Levinson-Durbin PARCOR coefficients $|k_m| < 1$ guarantee all poles of $1/A(z)$ are strictly inside the unit circle.